A method for similarity measure

The key point of this method is:
An image includes many features. In measuring similarity, some of the features do not contribute to the similarity, but rather introduce noise, and they therefore should be discarded. Based on this assumption, we introduce the idea of a similarity threshold, which functions as a filter to get rid of the noise features.

In this method, the similarity measure is a dynamic process, i.e. the number of similar feature elements varies, and for different pair of images, the similar channels may be different.

For example, there are three check-board images. One (A) is composed of small black and white squares. The other (B) includes red and yellow squares in the same size. The third one (C) also consists of black and white squares, but in very large size. Then we change the square size of C from very large to very small to compare their similarities.

1. If the square size of C is very large, we probably will say the A and B are more similar than A and C.
2. If the square size of C is near the size of B and A, either larger or smaller, we may say that A and C are much more similar than A and B.
3. If the square size of C is very small, we will say A and B are more similar than A and C again.

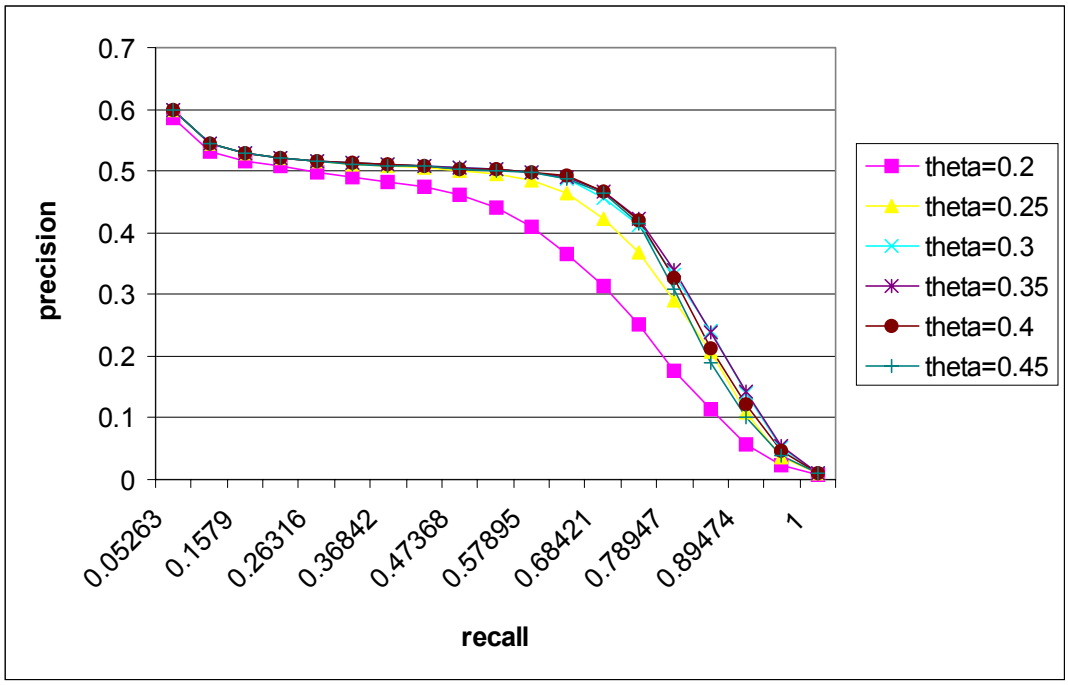
From this kind of phenomenon, we can see there is a threshold of the feature element distance between similar and dissimilar images. If we use probability to represent the distance, for instance, 0 is the same (very similar), and 1 is totally different, then the threshold between similar and dissimilar images may be 0.3 or so (we could get this threshold by training large image dataset). Feature distances within 0 and 0.3 contribute to the similarity, and those larger than 0.3 are taken as noise and should be discarded. Within 0 and the threshold, we get different numbers of similar feature elements for similar images, and also the channels contributing to the similarity are dynamically chosen.

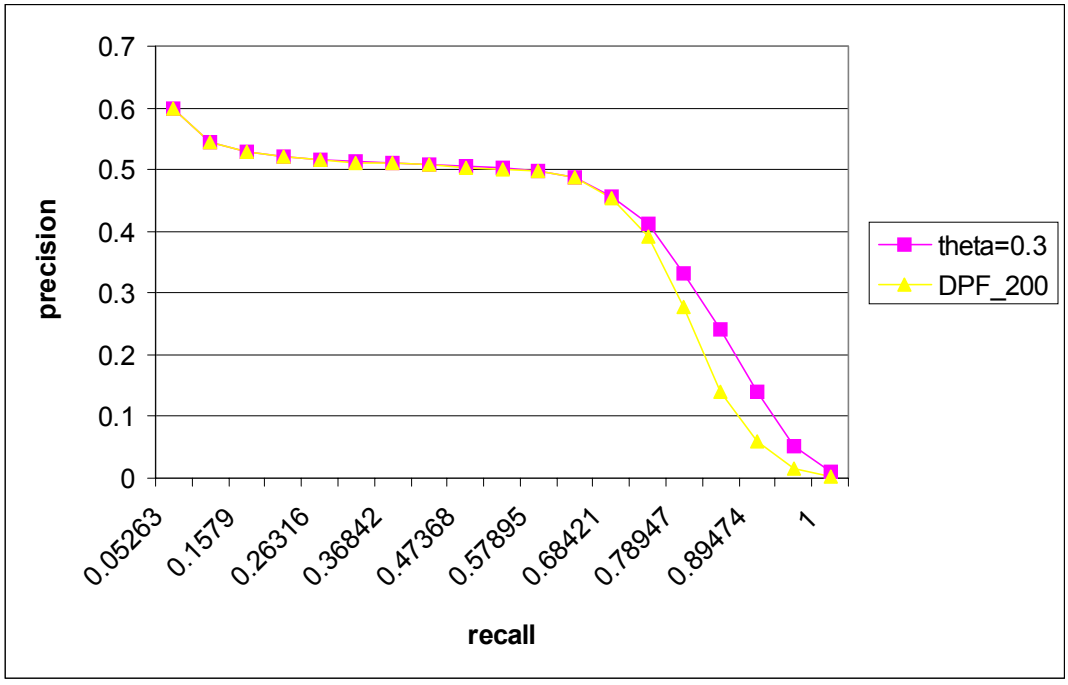
Then, we compute the similarity distance. Since the similar channels are unknown, I cluster them as one element by averaging them, and use this mean value as the distance. The results are very encouraging. This is based on the following idea: each image may include thousands of features, but we could only extract some of them, there are still lots of hidden features that could contribute to the similarity. By clustering the similar features as one representing element, we could reduce the possible bias impact.

I compared DPF with my method using both seeding and different kinds of transformed images as the query, such as Seeding, Crop_10%, Downsampling_90%, Changing Format, Framing_red, Framing_yellow, Rotation_180deg, Rotation_270deg, Rotation_90deg, Scaledown_2folds, Scaleup_5folds images. And I found my method is better than DPF in all these experiments.

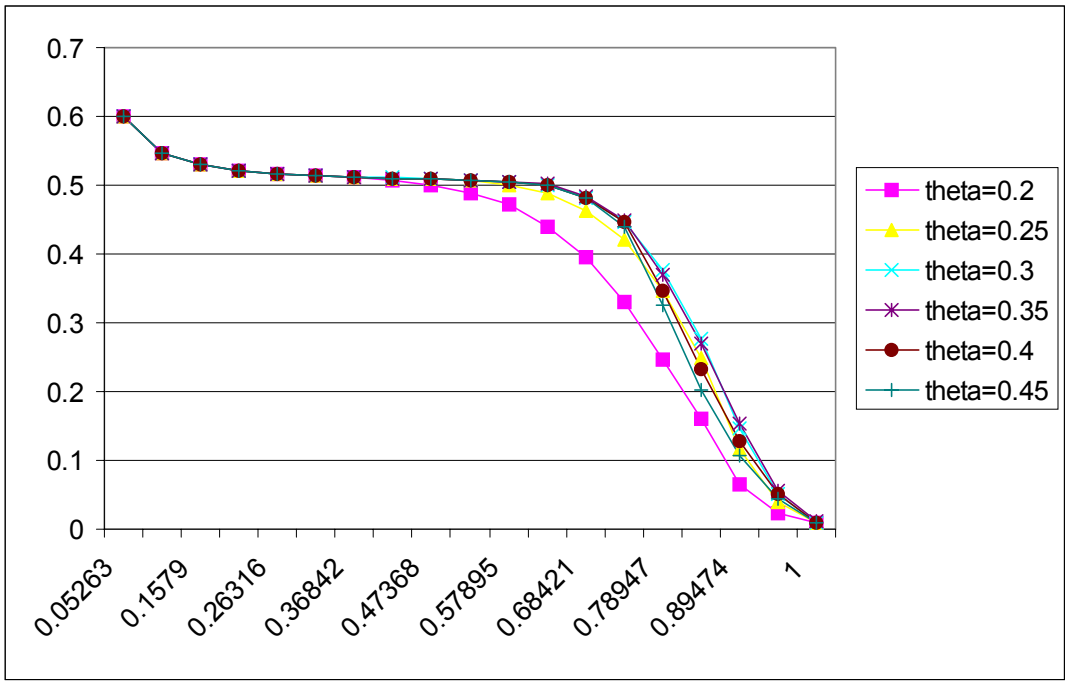
The results of the initial implementation of this method are included in the following pages. In each page, there are two figures for each experiment. The first figure is to find the threshold “theta” by training image dataset (for example, theta=0.3, i.e. the precision is the highest under the same recall rate). The second is the comparison between DPF and my method, both of which are set under the best parameters.

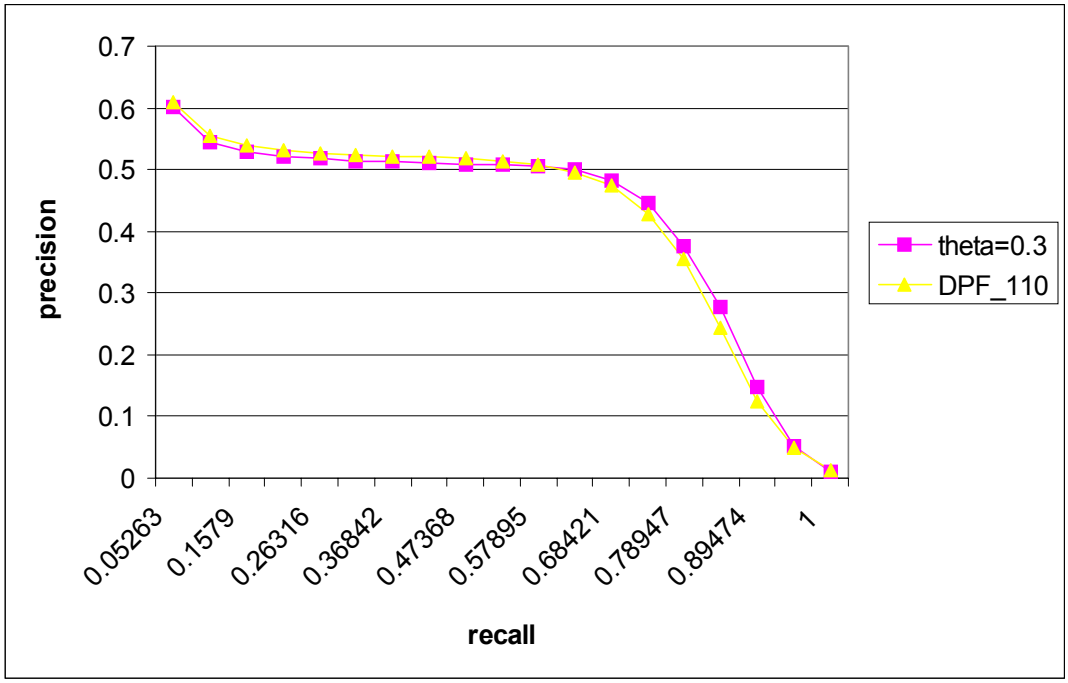
Crop_10


precision
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.2
theta=0.25
theta=0.3
theta=0.35
theta=0.4
theta=0.45

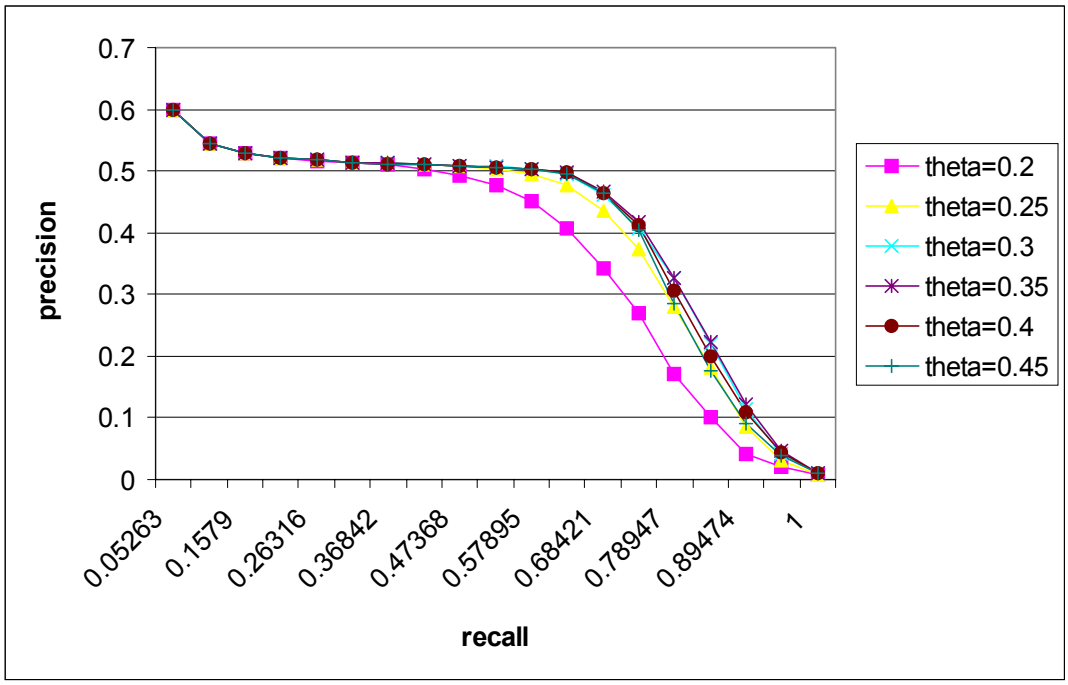

precision
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.3
DPF_200

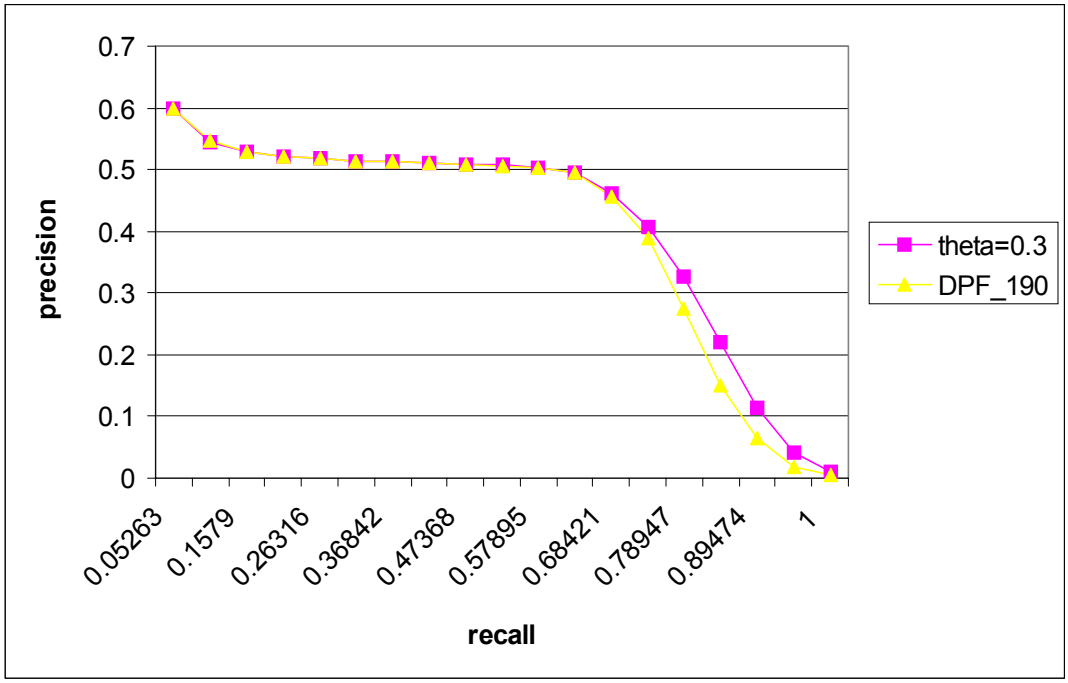
Downsample_90%



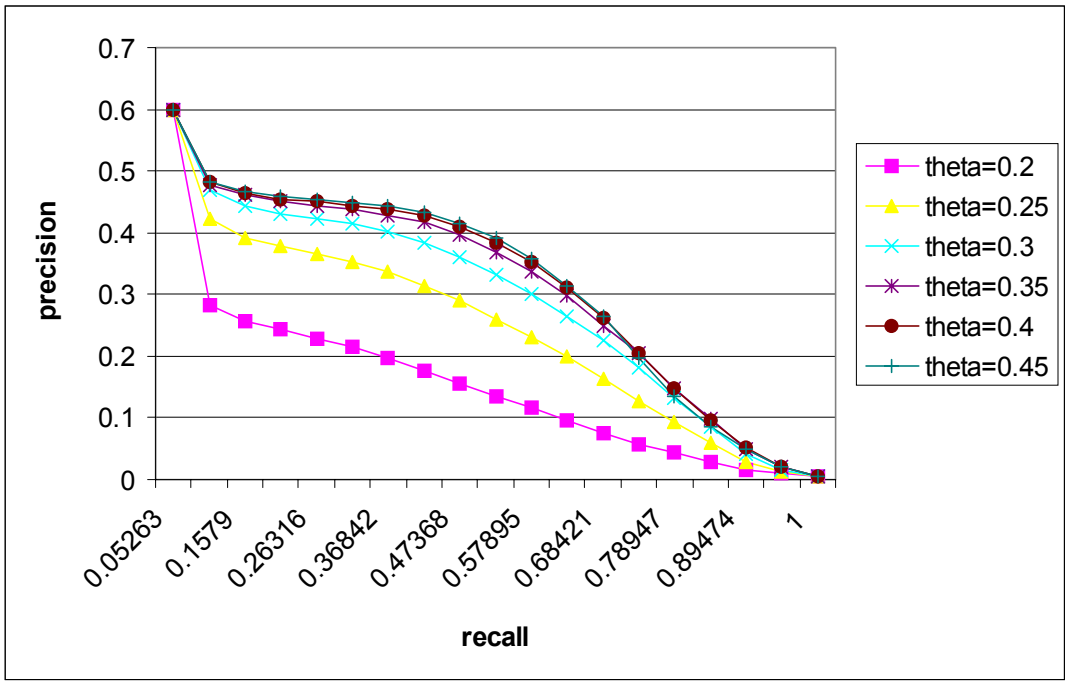


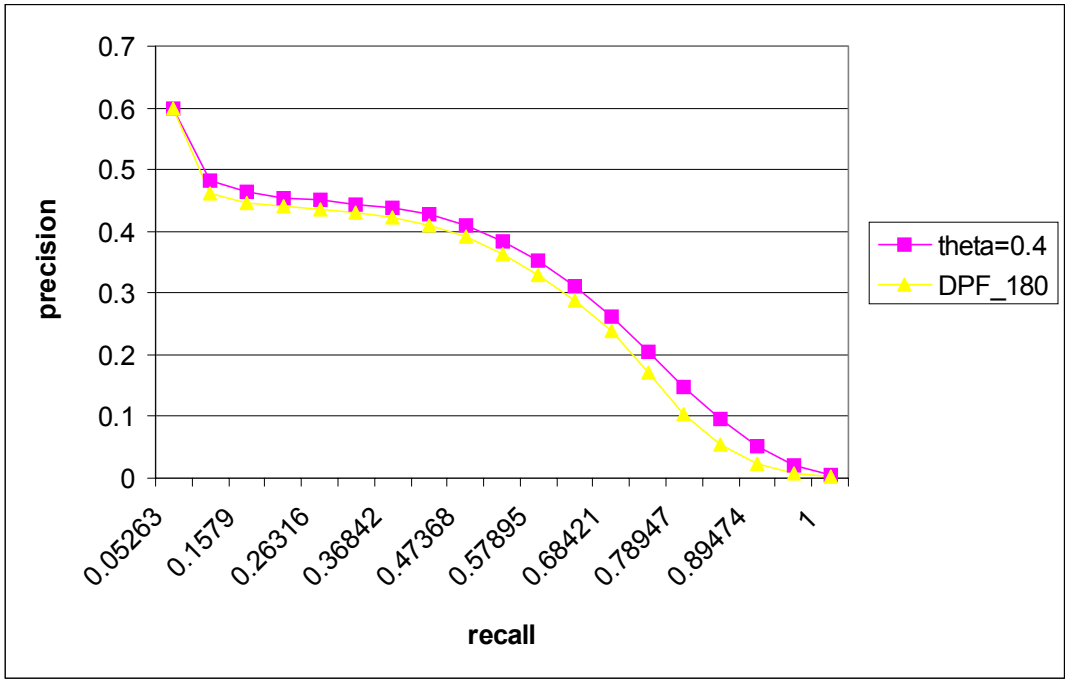
Format


0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.2
theta=0.25
theta=0.3
theta=0.35
theta=0.4
theta=0.45

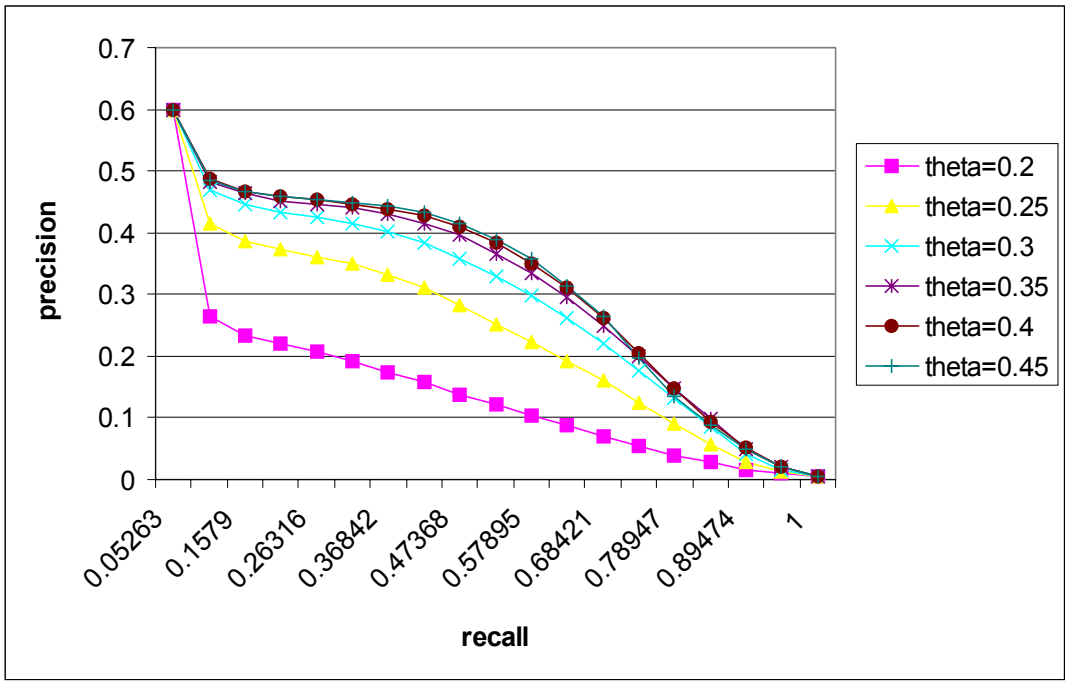

0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.3
DPF_190

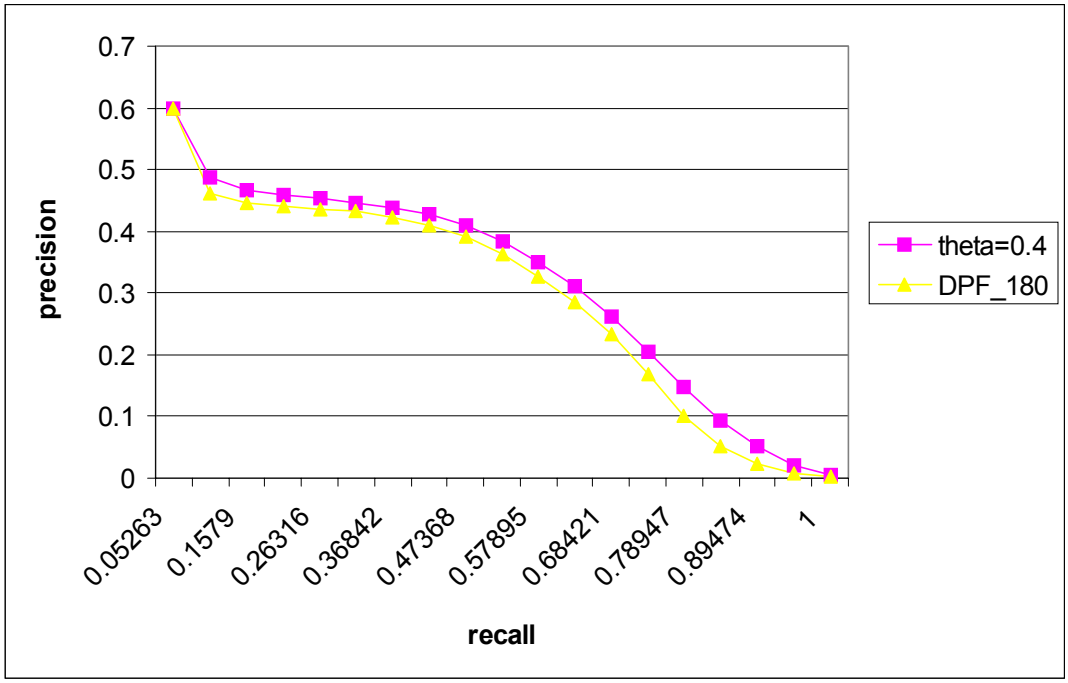
Frame_red


0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.2
theta=0.25
theta=0.3
theta=0.35
theta=0.4
theta=0.45

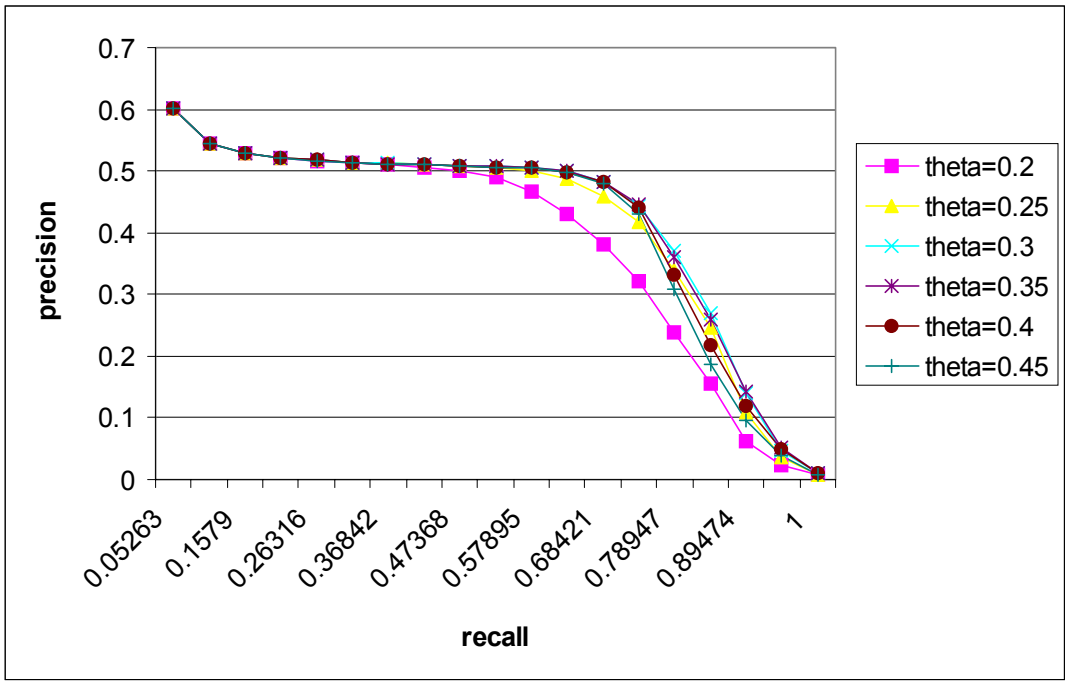

0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.4
DPF_180

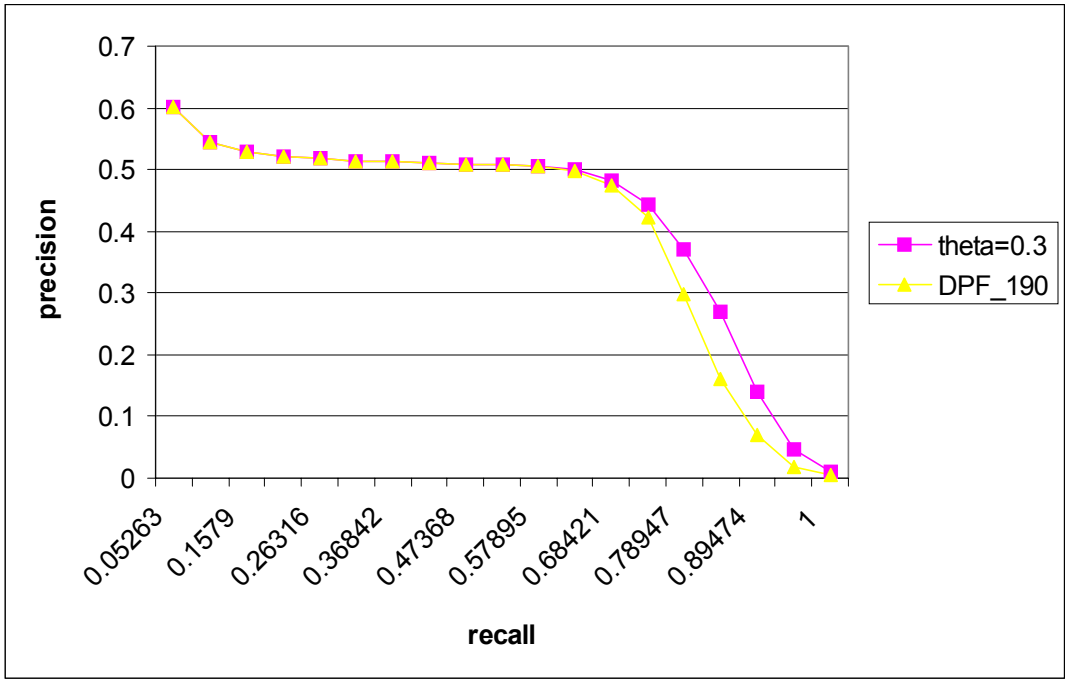
Frame_yellow



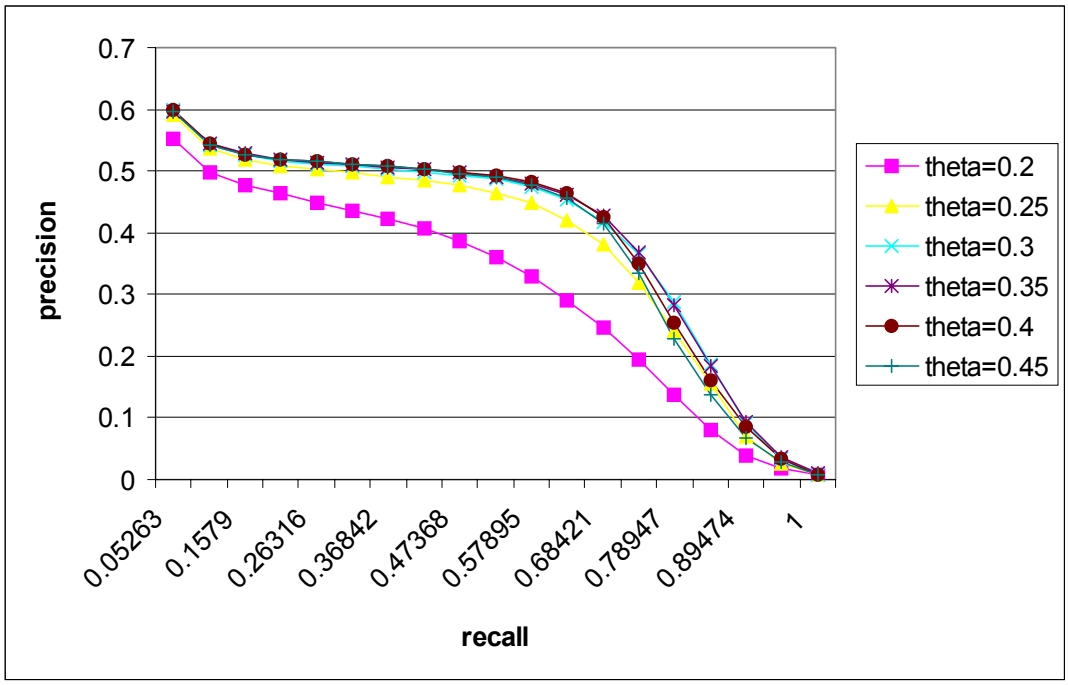


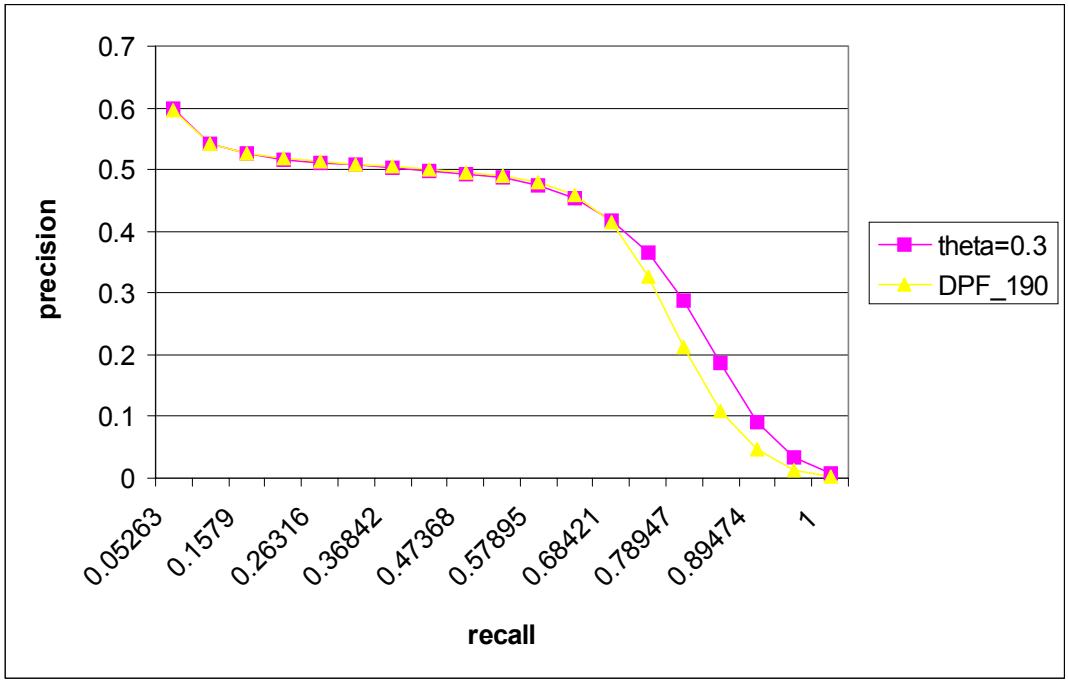
Rotation_180



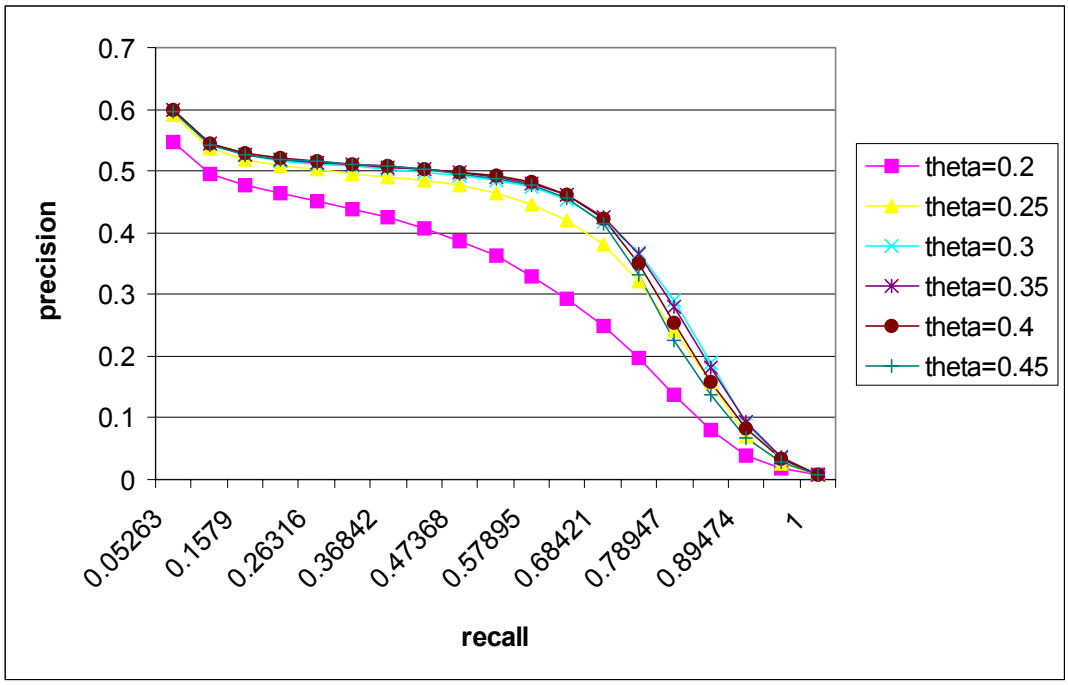


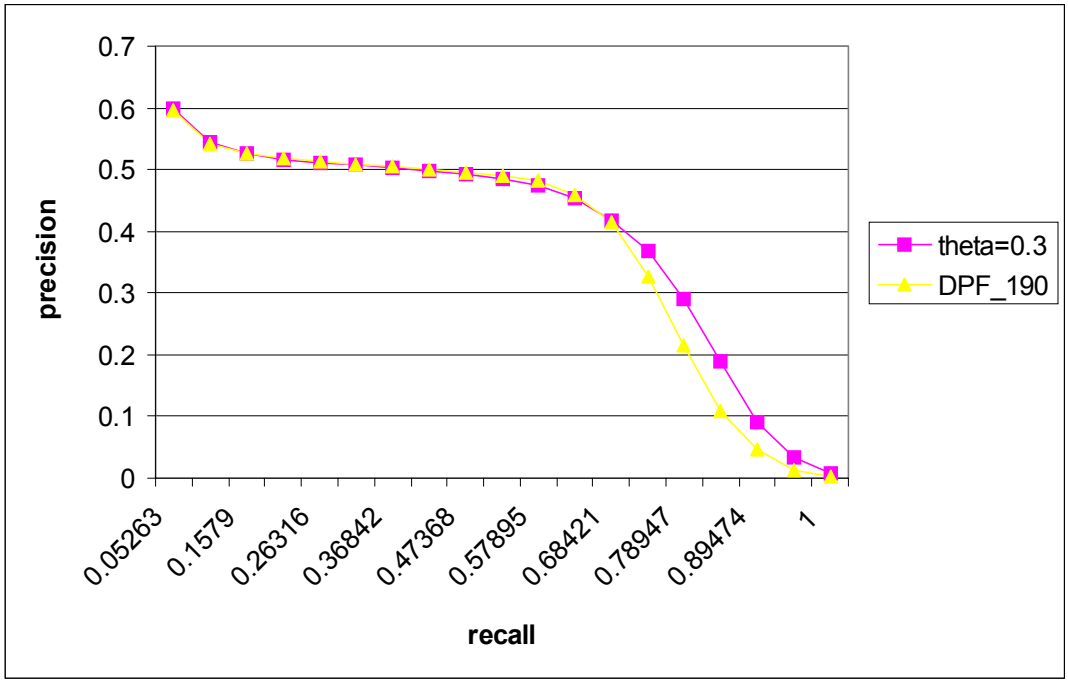
Rotation_270



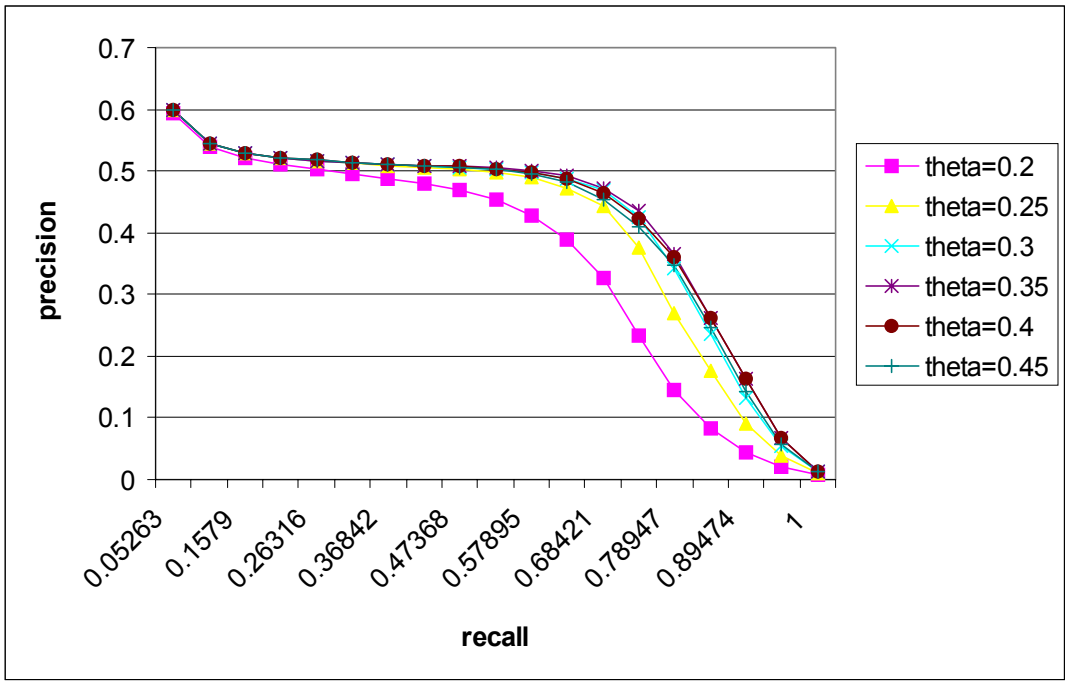


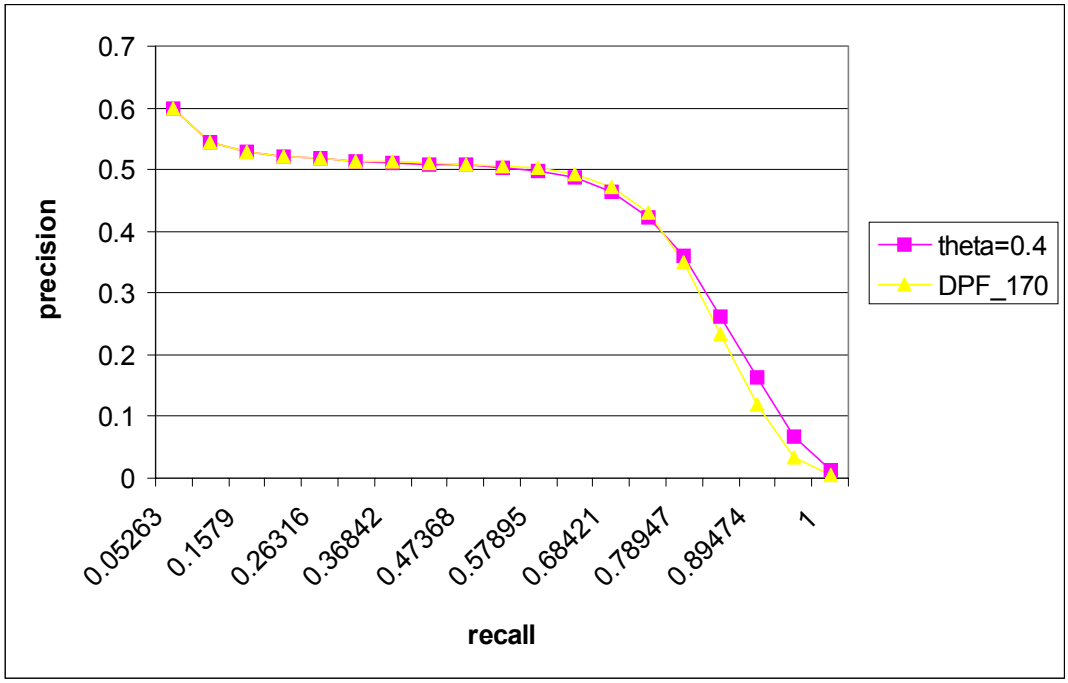
Rotation_90


0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.2
theta=0.25
theta=0.3
theta=0.35
theta=0.4
theta=0.45

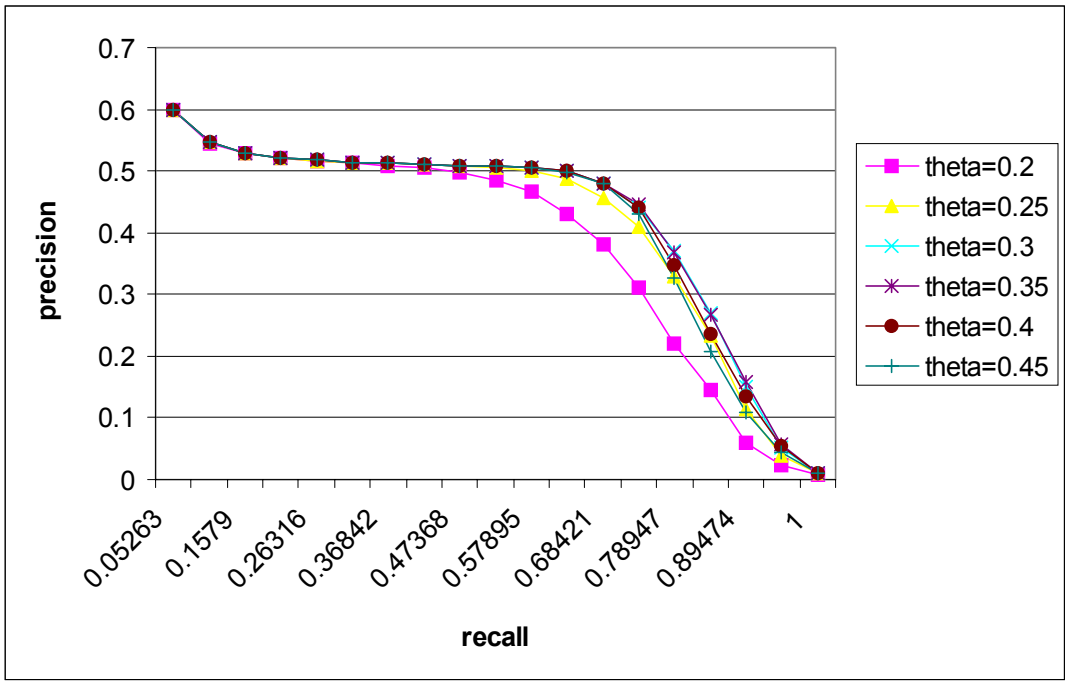

0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.3
DPF_190

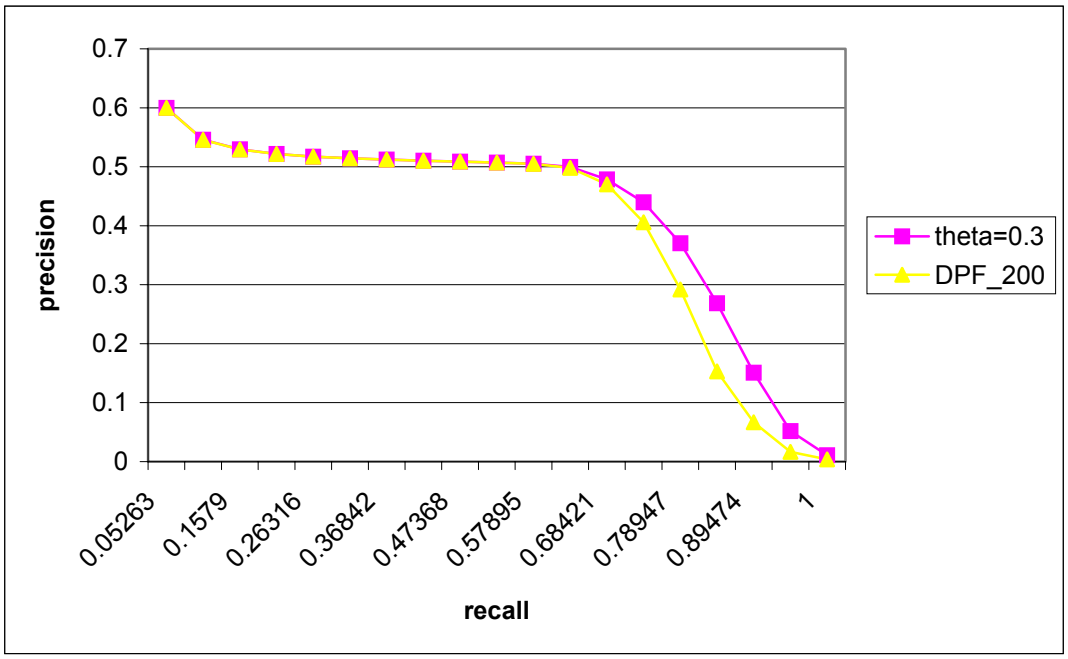
Scaledown_2


0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.2
theta=0.25
theta=0.3
theta=0.35
theta=0.4
theta=0.45

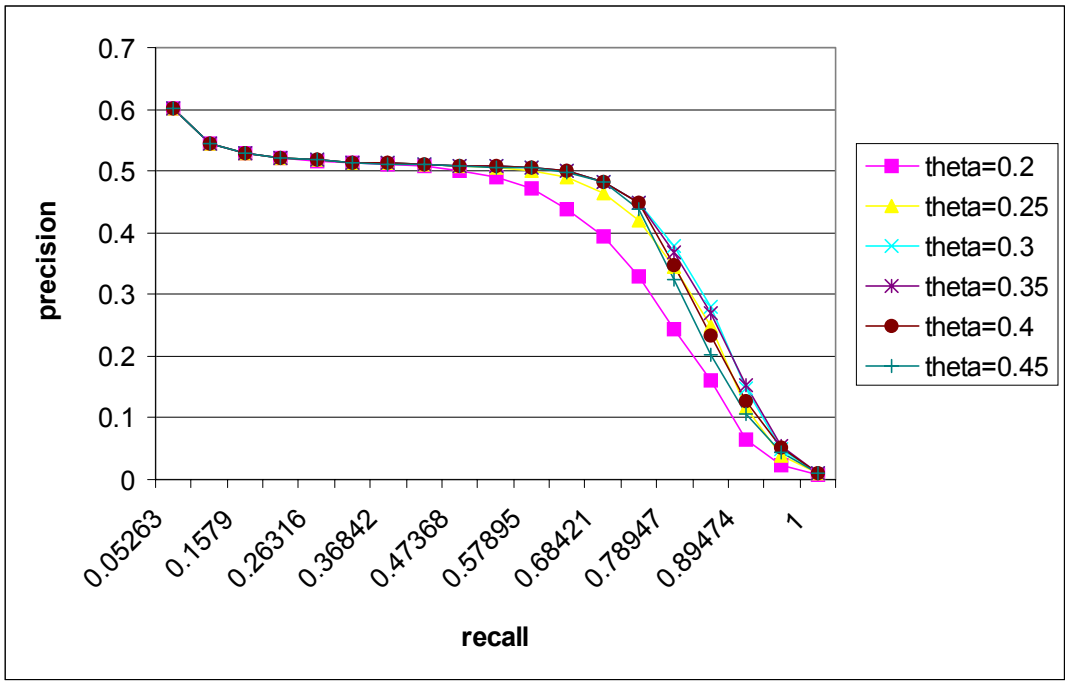

0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
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0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.4
DPF_170

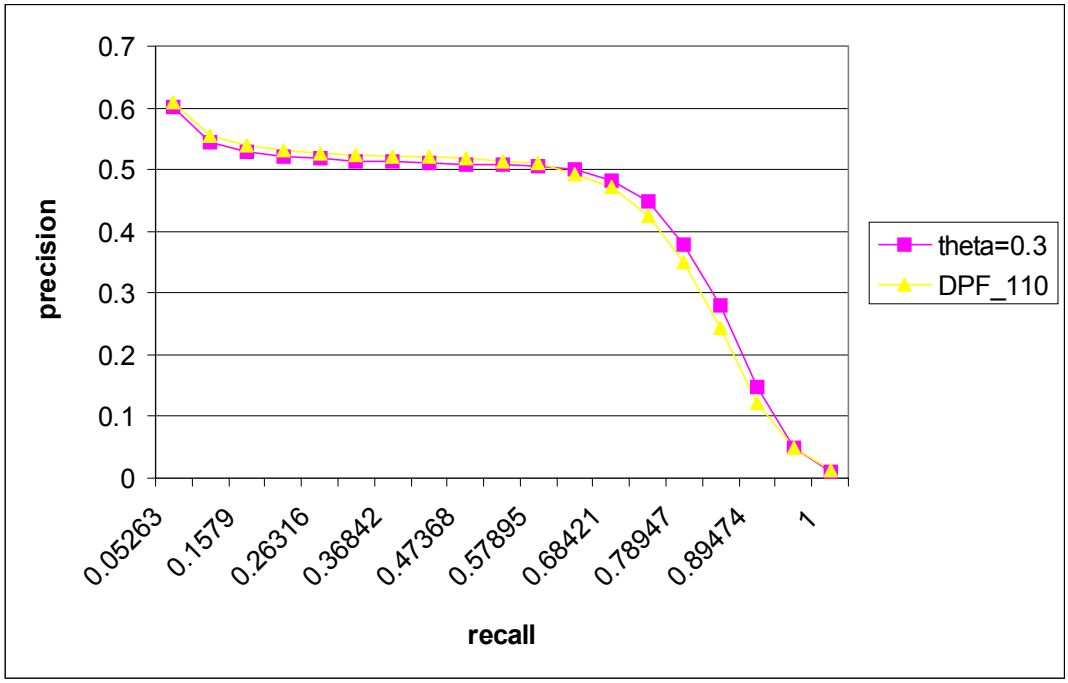
Scaleup_5


0.7
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0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.2
theta=0.25
theta=0.3
theta=0.35
theta=0.4
theta=0.45


0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.3
DPF_200

Seeds


0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.2
theta=0.25
theta=0.3
theta=0.35
theta=0.4
theta=0.45


0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
precision
0.05263
0.1579
0.26316
0.36842
0.47368
0.57895
0.68421
0.78947
0.89474
1
recall
theta=0.3
DPF_110